

SIDNEY-RICHLAND, MT, USA

WMO: 727687

Lat: 47.717N

Lon: 104.183W

Elev: 604

StdP: 94.28

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 94028

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.8	-24.9	-32.6	0.2	-25.0	-30.6	0.2	-23.1	13.3	-2.3	12.0	-2.6	2.4	190	0.652

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	34.5	20.0	32.6	19.6	30.9	19.2	22.2	30.7	21.1	29.6	20.0	28.3	4.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	15.0	26.8	17.9	13.9	25.3	17.2	13.2	24.3	68.3	31.0	63.7	29.5	59.6	28.2	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.9	10.6	9.1	-32.3	37.9	3.4	2.0	-34.7	39.4	-36.7	40.6	-38.6	41.7	-41.0	43.1		
(5)			-31.2	24.4	2.5	1.5	-33.1	25.5	-34.5	26.4	-35.9	27.2	-37.8	28.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.5	-9.1	-7.4	-0.7	6.6	12.4	17.9	22.0	20.8	15.1	7.0	-1.1	-6.9		
	DBStd	12.37	8.39	8.09	7.63	5.30	4.51	3.78	3.29	3.61	4.67	5.15	6.72	7.51		
	HDD10.0	2552	592	487	334	126	27	1	0	0	10	118	333	524		
	HDD18.3	4613	851	721	590	353	189	52	7	18	118	351	582	782		
	CDD10.0	1260	0	0	3	23	101	237	373	334	162	26	1	0		
	CDD18.3	280	0	0	0	1	6	38	121	93	21	1	0	0		
	CDH23.3	3917	0	0	2	28	139	553	1530	1246	386	31	0	0		
	CDH26.7	1673	0	0	0	7	34	190	711	572	154	6	0	0		
(14) Wind	WSAvg	4.0	4.1	4.0	4.2	4.5	4.4	4.0	3.4	3.4	3.6	4.1	3.9	4.0		
(15) Precipitation	PrecAvg	367	10	7	14	30	57	74	61	34	29	26	12	9		
	PrecMax	524	32	36	56	96	136	145	205	120	144	100	66	28		
	PrecMin	188	0	0	1	1	11	34	10	2	1	4	0	0		
	PrecStd	73	8	7	11	25	32	33	44	22	29	21	15	7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	12.7	22.2	27.4	30.1	34.6	37.7	37.3	34.1	27.3	17.9	10.1		
		MCWB	4.8	6.0	11.0	13.2	16.2	20.0	20.9	20.2	18.4	14.4	9.4	4.6		
	2%	DB	6.3	8.0	17.2	22.5	27.2	31.1	35.0	34.0	31.1	22.4	13.8	7.2		
		MCWB	2.7	3.6	8.5	11.6	15.1	19.3	20.7	19.2	17.8	12.6	6.9	2.9		
	5%	DB	3.9	5.3	12.9	19.1	24.1	28.7	32.7	32.3	27.7	18.7	11.1	4.6		
		MCWB	1.4	2.0	6.3	9.8	13.9	18.1	20.3	19.1	16.2	10.6	5.6	1.4		
	10%	DB	2.3	2.8	9.8	17.1	22.2	26.9	31.1	29.9	24.8	16.1	7.8	2.6		
		MCWB	0.4	0.5	4.9	8.7	13.0	17.1	19.9	18.4	14.9	9.2	3.7	0.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	6.6	11.1	13.9	18.6	22.4	24.6	23.0	20.4	15.7	10.1	5.1		
		MCDB	9.2	11.8	20.9	24.5	25.4	30.5	32.6	31.4	30.0	24.5	17.1	9.2		
	2%	WB	2.9	4.1	8.8	11.9	16.4	20.7	23.0	21.3	18.7	13.2	7.6	3.1		
		MCDB	6.1	8.0	16.1	21.3	24.7	28.4	31.1	30.6	28.2	21.0	13.3	6.7		
	5%	WB	1.5	2.3	6.8	10.5	15.0	19.3	21.7	20.2	17.2	11.3	5.6	1.7		
		MCDB	4.0	5.2	12.8	18.5	22.6	26.6	30.1	29.2	25.9	17.8	10.3	4.5		
	10%	WB	0.3	0.7	4.9	9.1	13.6	17.9	20.6	19.3	15.7	9.8	4.0	0.4		
		MCDB	2.3	2.9	9.3	16.0	20.6	25.1	28.8	27.9	23.6	15.5	7.6	2.6		
(35) Mean Daily Temperature Range	MDBR		10.6	11.1	12.3	13.7	13.7	13.6	14.9	15.7	15.5	13.1	11.6	10.5		
	5% DB	MCDBR	11.4	13.4	18.1	19.3	18.0	17.7	18.4	19.3	20.9	19.2	16.3	12.3		
		MCWBR	8.8	9.6	11.1	10.4	9.1	8.6	8.1	8.3	10.1	10.8	10.6	9.0		
	5% WB	MCDBR	11.2	12.8	17.7	17.6	15.9	15.8	16.0	16.7	18.5	17.6	15.3	12.1		
		MCWBR	8.8	9.3	11.0	9.9	8.6	8.3	7.8	7.8	9.2	10.2	10.2	9.1		
(40) Clear-Sky Solar Irradiance	taub		0.251	0.268	0.278	0.320	0.338	0.354	0.362	0.367	0.325	0.293	0.263	0.248		
	taud		2.488	2.486	2.542	2.425	2.436	2.430	2.432	2.389	2.480	2.559	2.534	2.506		
	Ebn at Noon		859	915	957	934	925	908	895	875	887	873	837	822		
	Edh at Noon		61	76	85	106	109	111	109	109	90	71	58	53		
(44) All-Sky Solar Radiation	RadAvg		1.53	2.53	3.75	4.91	5.85	6.51	6.97	5.71	4.20	2.66	1.63	1.15		
	RadStd		0.13	0.22	0.28	0.28	0.50	0.43	0.28	0.36	0.35	0.26	0.19	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page